

CUSTOMER MILESTONE BRIEF

MS #05 - Evidence-Grounded Graph Chat

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.5.0
Date	2026-08-10
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #5 - Evidence-Grounded Graph Chat converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Lets users ask plain-language questions about a legacy estate while requiring answers to cite retrieved graph evidence and state their limitations.

Why this matters to a customer

Lets users ask plain-language questions about a legacy estate while requiring answers to cite retrieved graph evidence and state their limitations.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added node-aware graph chat for who, what, where, when, why, how, impact, lineage, verification, and general explanation questions.
- Added intent-specific bounded retrieval and deterministic local answers with no external runtime dependencies.
- Added an optional OpenAI Responses API provider with strict structured output, server-side secret handling, disabled API storage, and a model override.
- Added claim citations, confidence rationale, limitations, supporting node and edge IDs, graph snapshot identity, and suggested follow-up questions to every answer.
- Added retrieval-time implementer/verifier isolation, audience-change history clearing, untrusted graph-data instructions, and rejection of citations outside the retrieved evidence package.
- Added a versioned answer schema and documented answer-quality and production-security contracts.
- Added six-question, prompt-injection, private-holdout, schema, provider, HTTP, and citation- validation tests.

How it differs from earlier work

MS #5 builds on MS #4 (Graph Explorer and Neo4j Projection). The earlier milestone established its own bounded capability; this milestone adds evidence-grounded graph chat while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use evidence-grounded graph chat as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.5.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity